McMurdo Station, Ross Island

BUILDING INFORMATION SHEET

Dorm

Location ID: B210

Two-story dormitory building housing up to 68 persons, constructed in McMurdo in 1994 to replace housing facilities at Williams Field III. First floor includes 14 two-person rooms, a communal bathroom, mechanical and electrical rooms, a lounge, a laundry, and a storage room. The second floor includes 20 two-person rooms, a communal bathroom, a storage room, and a mechanical room.



Photo 1: *Dorm*

General Structure Information

Usage	Dormitory
Occupancy Group	R-1 Dormitory
Maximum Occupancy	238
Construction Type	V, 1-Hour, combustible; steel-framed modular building with rigid foam insulation.
Year Constructed	1994
Dimensions	138' x 40'34'-6" x 12' extension at rear for mechanical room and toilet/showers

Floor Area

Floor	Area
Floor 1	5,934 sq ft85 Btu/hr oil-fired, forced-air furnace
Floor 2	5,934 sq ft125 Btu/hr oil-fired forced-air furnace
Total	11,868 sq ft

HVAC System

Heating	1st Floor:
Air Handling Units	Air to air heat exchanger provides building fresh air and combustion air to the first floor mechanical room. Bathroom exhaust

	tempers outside air.
Fuel Tank	Double-walled 1,000 gallon tank

Water and Sewer

Water Service	4-inch domestic supply
Sewer Service	6" stub off the 6" main
Connections	Sinks: Toilets: Urinals: Showers: Water Cooler: Ice Machine: Washing Machines: 40? lines
Water Heater	85 gallon oil-fired water heater

Electrical Service

Service Feeder	Feeder A
Transformer	500 kVA pad mount transformer (shared with B211)
Electrical Service	120/208V 3PH, 400A main breaker

Phone and LAN	40? lines

Fire System

Fire Panel Type	The building is equipped with a King-Fisher KFRTI-52 industrial radio and control system (IRACS) panel, pull stations, and alarm bells. The panel is located in the first floor electrical room.
Suppression Type	Pre-action sprinkler system connected to Firetrol pump and resevoir in Building 151 and equipped with Star, 3-inch, model D, electromagnetic deluge valve with air pressure supervision. Deluge valve releases upon low air pressure and heat detector actuation. Firetrol pump starts on water pressure loss in main trunk line to dorms.

Floorplans

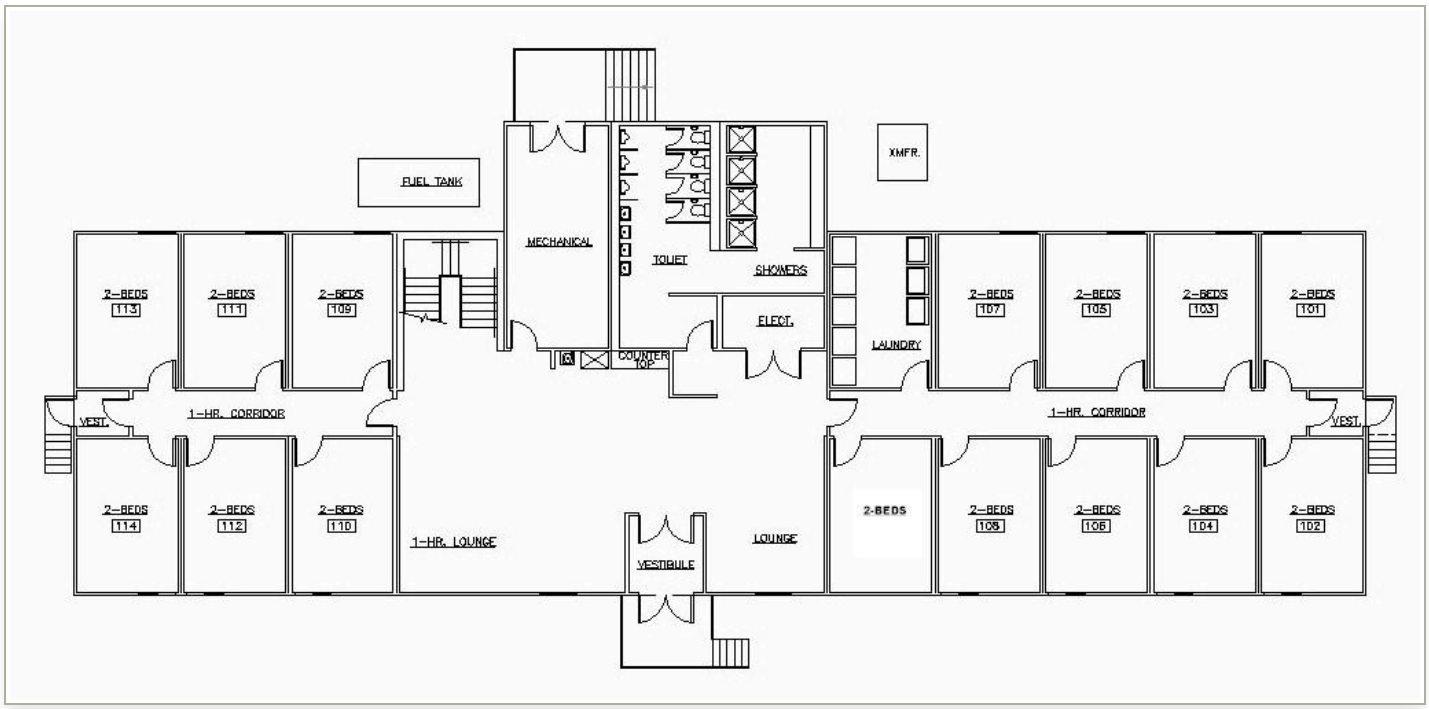


Figure 1: *Dorm*

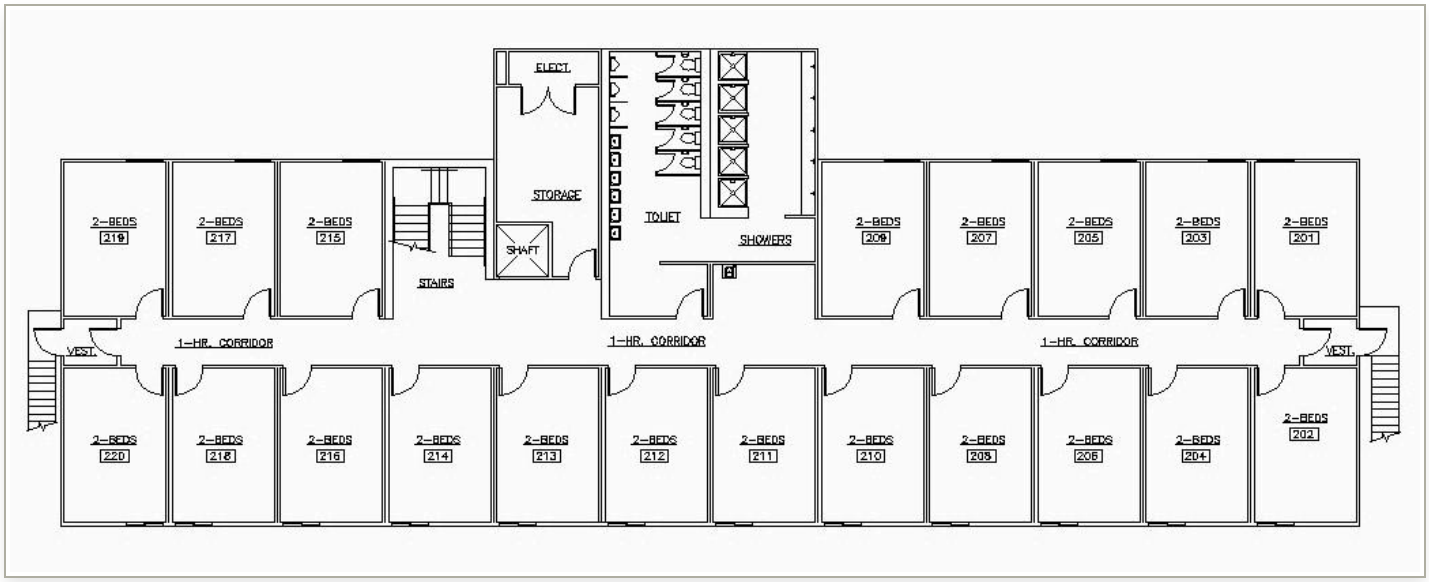


Figure 2: *Dorm*